

HOSNAIN RAFI

Open to Relocation to Japan / Remote

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SUMMARY

Full-Stack Software Engineer & Feature Lead with **4+ years of professional engineering experience** building scalable, real-time, and **Agentic AI-powered systems**. Specialized in multi-agent orchestration, LLM vision quality gates, high-throughput microservices, and distributed cloud applications. Proven track record leading fintech loan origination modules, automating complex data pipelines, and establishing rigorous quality assurance. Dedicated to disciplined Japanese workplace principles: proactive communication (**Hou-Ren-So**), continuous iteration (**Kaizen**), and uncompromising **Quality-First** engineering. Actively preparing for JLPT N5 and seeking long-term engineering placement in Japan.

TECHNICAL SKILLS

AI & Agentic Systems: Claude 3.5 Sonnet Vision API, OpenAI & Gemini APIs, Multi-Agent Swarms, Autonomous Task Orchestration, Tool / Function Calling, Automated UI Quality Gates (DesignGate), Prompt Engineering, RAG Architectures, Vector Embeddings

Languages: JavaScript (ES6+), TypeScript, C, C++, C# (.NET Core), Python, HTML5, CSS3/SCSS, SQL

Frontend: React 19, Angular, Next.js, Vue.js, Redux Toolkit, Pinia, Tailwind CSS, Responsive Design, State Management, DOM Optimization

Backend & APIs: Node.js, Express.js, RESTful APIs, Microservices Architecture, WebSocket / Socket.IO, WebRTC, Redis Pub/Sub, gRPC

Databases: PostgreSQL, MongoDB, MySQL, Redis (In-Memory Cache), SQLite, MariaDB, Prisma ORM, Mongoose

DevOps & Tooling: Docker, Linux (Ubuntu/CentOS), Git, GitHub, Bitbucket, Vercel, Netlify, Web Scraping (BeautifulSoup/Puppeteer), Vite

Testing & QA: Playwright E2E Testing, Jest Unit Testing, Postman API Automation, Visual Regression AI Analysis, CI/CD Automated Pipelines

EXPERIENCE

Feature Lead & Full-Stack Software Engineer | Mortgage Magic Ltd

Nov 2021 – Present

Remote for a London-based fintech enterprise | Angular, Node.js, Express, PostgreSQL, Redis, Docker, Playwright

- **Architecture & Engineering Leadership:** Led end-to-end development of critical UK loan origination workflows, broker portals, and customer application modules serving financial institutions.
- **High-Performance Distributed Backend:** Architected decoupled REST microservices in Node.js/Express with PostgreSQL and Redis caching, cutting average API response times by 35% across high-concurrency loan pipelines.
- **AI & Automation Pipelines:** Spearheaded the development of automated rate-scraping engines and criteria ingestion pipelines, integrating intelligent data extraction to reduce manual advisor data-entry effort by 60%.
- **Quality Gates & Kaizen Methodology:** Enforced rigorous code review standards, Playwright end-to-end regression suites, and automated linting, achieving zero critical production regressions across consecutive release cycles.
- **Cross-Functional Collaboration (Hou-Ren-So):** Maintained disciplined daily reporting, sprint retrospectives, and transparent technical documentation across distributed engineering, product, and QA stakeholders.
- **Mentorship & Standards:** Guided junior developers in TypeScript best practices, scalable component design, and git workflows.

Full-Stack Software Engineer | Freelance & Contract Engineering

Jan 2020 – Oct 2021

Global Remote | React, Node.js, TypeScript, MongoDB, MySQL, Socket.IO, Docker

- **Full-Stack Product Delivery:** Engineered and shipped 8+ responsive web applications and dashboards for international clients, delivering modular React frontends and robust Node.js backend services.
- **Real-Time Systems & Event Pipelines:** Designed event-driven web applications featuring WebSocket streaming, live notification feeds, and reactive dashboard state synchronization.
- **Client Relationship & Quality Delivery:** Managed full software lifecycles from requirement gathering and UI/UX prototyping to deployment on cloud infrastructure (Docker, Linux VPS, Vercel), maintaining a 100% on-time delivery record.

PROJECTS

DesignGate — Automated UI Quality Gate with Vision LLM

[Live Demo](#) | [GitHub](#) | 2025

Tech: Playwright Headless, Claude 3.5 Sonnet Vision API, Node.js, TypeScript, Next.js, Tailwind CSS

- Built an automated QA quality gate eliminating subjective UI code reviews by combining automated headless browser evidence collection with multi-modal AI evaluation.
- **Automated Evidence Pipeline:** Captures synchronized desktop and mobile viewports, full-page snapshots, and DOM hierarchy using Playwright.
- **Vision LLM Quality Grading:** Integrates Claude 3.5 Sonnet Vision to evaluate design fidelity, layout drift, typography consistency, contrast ratios, and responsive anomalies against baseline specifications.
- **CI/CD Readiness:** Generates machine-readable JSON pass/fail verdicts and interactive visual audit reports, enabling zero-human-touch visual regression gating in deployment pipelines.

Agent-Swarm — Autonomous Multi-Agent CLI Orchestrator

[GitHub Repo](#) | 2025

Tech: Python, OpenAI / Claude LLM APIs, Asyncio, Multi-Agent Architecture, CLI Tooling

- Engineered a distributed multi-agent swarm architecture for developer terminal assistants to execute complex, multi-stage software engineering workflows autonomously.
- **Hierarchical Agent Delegation:** Implemented specialized sub-agent roles (Planner, Coder, Reviewer, and Test Runner) communicating via structured message buses and shared context memory.
- **Self-Correction & Tool Execution:** Agents autonomously execute bash scripts, inspect source code trees, parse compiler/linter errors, and iterate on bug fixes until acceptance criteria are satisfied.

Urban Grill — Real-Time Operations & Restaurant Platform

[Live App](#) | [GitHub Repo](#) | 2025

Tech: React 19, Node.js, Express, MongoDB, Socket.IO, Tailwind CSS, Vercel

- Developed a full-stack restaurant operations platform with dual customer ordering and live kitchen administrative management.
- **Real-Time Kitchen Dispatch:** Engineered low-latency bidirectional WebSocket (Socket.IO) pipelines synchronizing orders instantly between customer mobile devices and kitchen display terminals.
- **Role-Based Operational Workflows:** Implemented JWT authentication and fine-grained access control across customer, kitchen staff, and executive admin portals with live inventory tracking.

EDUCATION

B.Sc. (Eng.) in Information and Communication Technology | Mawlana Bhashani Science and Technology University (MBSTU)

2019 – 2023

Tangail, Bangladesh • Rigorous 4-year engineering curriculum covering Data Structures & Algorithms, Distributed Systems, Software Engineering, Database Systems, Artificial Intelligence, and Computer Networks.

LANGUAGES & PROFESSIONAL PRINCIPLES

English: Business Level — Fluent in technical architecture discussions, sprint documentation, and global team collaboration.

Japanese: Basic / Conversational — JLPT N5 preparation in progress (daily study of technical terminology and conversation).

Work Principles: Active practitioner of Hou-Ren-So (proactive reporting, contact, and consultation), Kaizen (continuous iterative improvement), and Quality-First engineering standards.